# SIMATS ENGINEERING

## ASSESSMENT TOOL 2

**COURSE NAME:** **CSA16 COURSE CODE: Data Warehousing and Data Mining**

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**COURSE OUTCOME COVERED**

**CO2:** Apply suitable Data Preprocessing techniques like Data cleaning, Data intergration, Data transformation and data Reduction, for effective data preparation (BL3, BL4)

Assessment Tool Weightage: 50%

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**QUESTIONS: 5** **Total Marks:50**

Annexure A

SENARIO BASED QUESTIONS – To be Implemented in Weka

| ***Criteria*** | Understanding of Scenario ***(2)*** | Identification of Key Issues ***(2)*** | Application of Concepts ***(2.5)*** | Decision Making ***(2)*** | Clarity of Response ***(1.5)*** | ***Total(10)*** |
|---|---|---|---|---|---|---|
| *Weightage* | *20%* | 20% | 25% | 20% | 15% | *100%* |
| *Q1* | | | | | | |
| *Q2* | | | | | | |
| *Q3* | | | | | | |
| *Q4* | | | | | | |
| *Q5* | | | | | | |

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***Code of Conduct:***

*I M. Pranay (192525382) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:*** *M. Pranay*

**RUBRICS FOR CALCULATION:**

| Criteria | Weightage (%) | Level 4 (Exemplary) | Level 3 (Proficient) | Level 2 (Developing) | Level 1 (Beginning) |
|---|---|---|---|---|---|
| Understanding of Scenario | 20% | Demonstrates clear and complete understanding of the scenario and context | Shows good understanding with minor gaps | Partial understanding; misses key aspects | Misinterprets or fails to understand the scenario |
| Identification of Key Issues | 20% | Accurately identifies all relevant issues and factors involved | Identifies most key issues with minor omissions | Identifies limited issues; some irrelevant points | Unable to identify key issues |
| Application of Concepts | 25% | Applies appropriate concepts effectively to address the scenario | Applies concepts with minor errors | Limited or partially correct application | Unable to apply relevant concepts |
| Decision Making | 20% | Makes well-reasoned, logical decisions with strong rationale | Makes reasonable decisions with some justification | Makes weak decisions with limited justification | No clear decisions made or rationale provided |
| Clarity of Response | 15% | Response is clear, structured, and logically presented | Generally clear with minor issues | Somewhat unclear or poorly structured | Disorganized and difficult to understand |

**Annexure A**

Scenario based Questions

## Q1. Data Preprocessing in Retail Analytics ***(L3, L4)***

A **retail analytics company** is preparing customer purchase data for building a recommendation system. The dataset contains missing values, inconsistent categorical formats, and extreme values due to data collected from multiple stores and manual entries.

Sample Dataset

| Customer_ID | Age | Gender | Monthly_Spend (₹K) | Purchase_Frequency | Loyalty_Level |
|---|---|---|---|---|---|
| C1 | 23 | M | 20 | 5 | Low |
| C2 | NaN | Male | 35 | 8 | Medium |
| C3 | 31 | F | NaN | 10 | High |
| C4 | 105 | Female | 500 | 50 | High |
| C5 | 27 | male | 25 | NaN | Low |
| C6 | 40 | FEMALE | 45 | 12 | Medium |
| C7 | 36 | M | 38 | 9 | Low |
| C8 | 42 | Female | 300 | 45 | High |

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Tasks

### A. Handling Missing Values

i. Identify missing values in **Age, Monthly_Spend, and Purchase_Frequency**
ii. Apply at least **two imputation techniques** (e.g., Median Imputation, k-NN Imputation)
iii. Compute the imputed values and justify the selected method

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### B. Outlier Detection and Treatment

i. Detect outliers in **Age and Monthly_Spend** using the **Z-score method**
ii. Identify records containing extreme values (e.g., Age = 105, Spend = 500)
iii. Apply suitable treatment methods (**capping / transformation / removal**) and justify

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C. Categorical Data Standardization

i. Identify inconsistencies in **Gender** (M, Male, male, FEMALE, etc.)
ii. Convert all entries into a standardized format (**Male / Female**)

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Q2. Covariance & Data Reduction in E-commerce Analytics *(L4, L5)*

An **e-commerce platform** is building a predictive model to classify high-value customers. The dataset includes several correlated financial and behavioral attributes, leading to redundancy and increased computational cost.

Sample Dataset

| Cust_ID | Age | Income (₹K) | Purchase_Value (₹K) | Credit_Score | Orders | Avg_Balance (₹K) | EMI (₹) | Customer_Class |
|---|---|---|---|---|---|---|---|---|
| U1 | 24 | 40 | 100 | 650 | 15 | 12 | 5000 | Low |
| U2 | 34 | 60 | 150 | 700 | 20 | 18 | 7000 | Medium |
| U3 | 44 | 85 | 210 | 720 | 28 | 25 | 9000 | Medium |
| U4 | 52 | 120 | 320 | 800 | 35 | 40 | 15000 | High |
| U5 | 29 | 45 | 110 | 680 | 18 | 14 | 5500 | Low |
| U6 | 39 | 75 | 180 | 710 | 24 | 22 | 8500 | Medium |
| U7 | 56 | 150 | 360 | 820 | 40 | 50 | 18000 | High |
| U8 | 37 | 70 | 160 | 705 | 22 | 20 | 8000 | Medium |

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Tasks

A. Covariance Analysis

i. Compute covariance between **Income and Purchase_Value** (show steps or partial calculation)
ii. Interpret whether the relationship is **positive or negative**
iii. Explain how covariance helps in identifying feature relationships

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B. Feature Selection

i. Identify redundant attributes using **correlation or domain knowledge**
ii. Select an optimal subset of features for modeling
iii. Justify your selection

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C. Data Transformation

i. Standardize the dataset (excluding **Cust_ID and Customer_Class**)
ii. Explain why normalization/standardization is required

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## Q3. Data Transformation & Discretization in Education Analytics ***(L3, L4, L5)***

An **education analytics team** is developing a model to predict student performance. Continuous variables such as study hours and test scores need to be discretized to improve interpretability and decision-making.

Sample Dataset

| Student_ID | Study_Hours | Test_Score | Attendance (%) | Performance |
|---|---|---|---|---|
| S1 | 2 | 40 | 60 | Poor |
| S2 | 3 | 50 | 65 | Average |
| S3 | 4 | 55 | 70 | Average |
| S4 | 5 | 65 | 75 | Good |
| S5 | 6 | 75 | 80 | Good |
| S6 | 7 | 85 | 85 | Good |
| S7 | 8 | 90 | 88 | Excellent |
| S8 | 9 | 95 | 90 | Excellent |
| S9 | 10 | 98 | 92 | Excellent |
| S10 | 11 | 100 | 95 | Excellent |

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Tasks

A. Equal-Width Binning

i. Apply **3 bins** for Study_Hours and Test_Score
ii. Calculate bin width and define intervals

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B. Equal-Frequency Binning

i. Divide data into **3 bins with equal records**
ii. Assign values accordingly

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C. Concept Hierarchy Construction

- Study_Hours → Low / Medium / High
- Test_Score → Poor / Average / Excellent

**4.** Use the two methods below to normalize the following group of data:
200, 300,400, 600, 1000
(a) min-max normalization by setting min = 0 and max = 1
(b) z-score normalization

## 5. Data Transformation in Online Shopping Behavior Analysis ***(L3, L4, L5)***

An **e-commerce company** is analyzing customer engagement by measuring the **time spent (in seconds)** on product pages. The duration varies significantly—from a few seconds to more than an hour—resulting in highly skewed data.

To improve the effectiveness of **customer segmentation using clustering techniques**, the dataset must be properly transformed and normalized.

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### Sample Dataset: Visit Duration

| Customer_ID | Visit_Duration (seconds) |
|---|---|
| C1 | 15 |
| C2 | 30 |
| C3 | 45 |
| C4 | 60 |
| C5 | 120 |
| C6 | 300 |
| C7 | 600 |
| C8 | 900 |

| Customer_ID | Visit_Duration (seconds) |
|---|---|
| C9 | 1800 |
| C10 | 3600 |

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## ☑ Tasks

### A. Equal-Frequency Binning

i. Divide the dataset into **3 bins** with approximately equal number of records
ii. Assign each value to its respective bin
iii. Analyze how effectively this method groups short and long visit durations

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### B. Z-Score Normalization

i. Compute the **mean and standard deviation** of visit duration
ii. Apply Z-score normalization:

$$Z = \frac{X - \mu}{\sigma}$$

iii. Discuss how this method handles **extreme values (e.g., 1800, 3600 seconds)**

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### C. Min-Max Normalization

i. Apply Min-Max normalization using range **[0,1]**:

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

ii. Transform all values accordingly
iii. Compare results with Z-score normalization in terms of **scaling and sensitivity to outliers**

**ANSWER**

## 1. Covariance Analysis in Health Insurance

Given data:

| Premium increase (xi) | Claims (yi) |
|---|---|
| 3.2 | 15 |
| 4.0 | 18 |
| 5.5 | 25 |
| 4.8 | 20 |

Mean of x: $\bar{x} = (3.2 + 4.0 + 5.5 + 4.8) / 4 = 17.5 / 4 = 4.375$.

Mean of y: $\bar{y} = (15 + 18 + 25 + 20) / 4 = 78 / 4 = 19.5$.

Population covariance: $Cov(x,y) = \Sigma[(xi - \bar{x})(yi - \bar{y})] / n$.

| xi | yi | $xi - \bar{x}$ | $yi - \bar{y}$ | Product |
|---|---|---|---|---|
| 3.2 | 15 | −1.175 | −4.5 | 5.2875 |
| 4.0 | 18 | −0.375 | −1.5 | 0.5625 |
| 5.5 | 25 | 1.125 | 5.5 | 6.1875 |
| 4.8 | 20 | 0.425 | 0.5 | 0.2125 |
| Total | | | | 12.25 |

Cov(x,y) = 12.25 / 4 = 3.0625.

**Interpretation:** Covariance is positive, so premium increase and claim frequency have a positive relationship. As premium increase rises, claim frequency tends to increase.

**Note:** If sample covariance is required, divide by n − 1 = 3; the result is 4.0833. The relationship remains positive.

## 2. Data Smoothing Using Equal-Frequency Binning

Given 27 sorted systolic blood pressure readings. Divide the data into 3 equal-frequency bins of 9 values each and smooth each bin using its mean.

Bin 1: 110, 112, 115, 115, 118, 120, 120, 122, 124. Mean = 1056 / 9 = 117.33. Smoothed values: 117.33 repeated 9 times.

Bin 2: 124, 126, 126, 126, 126, 130, 135, 135, 138. Mean = 1166 / 9 = 129.56. Smoothed values: 129.56 repeated 9 times.

Bin 3: 138, 138, 138, 140, 145, 150, 152, 160, 180. Mean = 1341 / 9 = 149. Smoothed values: 149 repeated 9 times.

Result: Equal-frequency binning reduces minor fluctuations and makes the overall trend easier to analyze.

## 3. Equal-Width Binning for Risk Categorization

Given sorted BMI values: 18.5, 19.2, 21.8, 24.9, 25.5, 26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4, 33.1, 33.6, 34.7, 38.2, 39.5.

Range = 39.5 − 18.5 = 21. Assuming 3 equal-width bins, bin width = 21 / 3 = 7.

| Bin | BMI values | Smoothed values (nearest boundary) |
|---|---|---|
| 1 | 18.5, 19.2, 21.8, 24.9, 25.5 | 18.5, 18.5, 18.5, 25.5, 25.5 |
| 2 | 26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4 | 25.5, 25.5, 25.5, 32.5, 32.5, 32.5, 32.5, 32.5 |
| 3 | 33.1, 33.6, 34.7, 38.2, 39.5 | 32.5, 32.5, 32.5, 39.5, 39.5 |

**Result:** Equal-width binning divides BMI values into intervals of equal width and replaces values with the nearest bin boundary.

**Note:** The question does not specify the number of bins; this answer assumes 3 bins.

## 4. Data Reduction Using Bin Medians (Medical Supply Costs)

Sorted costs: 120, 140, 145, 145, 148, 150, 150, 150, 155, 155, 155, 155, 160, 160, 165, 170, 170, 175, 175, 175, 175, 180, 190, 200, 210, 230, 300.

There are 27 values. Divide into 3 equal-frequency bins of 9 values each.

Bin 1: 120, 140, 145, 145, 148, 150, 150, 150, 155. Median = ₹148. Replace all values in the bin with ₹148.

Bin 2: 155, 155, 155, 155, 160, 160, 165, 170, 170. Median = ₹160. Replace all values in the bin with ₹160.

Bin 3: 175, 175, 175, 175, 180, 190, 200, 210, 230. Median = ₹180. Replace all values in the bin with ₹180.

Result: Bin-median smoothing reduces variation in medical supply costs and simplifies the data for mining.

**Note:** The supplied data contains 27 values; the last value ₹300 is omitted from the three 9-value bins above. If all values must be included, the bins need to be adjusted.

## 5. Standardization of Patient Age Data

Given ages: 18, 22, 25, 42, 28, 43, 33, 35, 56, 28.

Mean: $\bar{x} = 330 / 10 = 33$ years.

Absolute deviations from the mean:

| Age (xi) | $\lvert xi - 33 \rvert$ |
|---|---|
| 18 | 15 |
| 22 | 11 |
| 25 | 8 |
| 42 | 9 |
| 28 | 5 |
| 43 | 10 |
| 33 | 0 |
| 35 | 2 |
| 56 | 23 |
| 28 | 5 |
| Total | 88 |

$MAD = \Sigma|xi - \bar{x}| / n = 88 / 10 = 8.8$ years.

Population standard deviation for z-scores: $\Sigma(xi - \bar{x})^2 = 1174$; $\sigma = \sqrt{(1174 / 10)} = \sqrt{117.4} \approx 10.835$.

| Age | Calculation | Z-score |
|---|---|---|
| 18 | $(18 - 33) / 10.835$ | −1.38 |
| 22 | $(22 - 33) / 10.835$ | −1.02 |
| 25 | $(25 - 33) / 10.835$ | −0.74 |
| 42 | $(42 - 33) / 10.835$ | 0.83 |

Final answers: Mean age = 33 years; MAD = 8.8 years; first four z-scores = −1.38, −1.02, −0.74, 0.83.

Interpretation: Negative z-scores indicate ages below the mean; a positive z-score indicates an age above the mean.